Amazon Sales Data Cleaning & SQL Analysis Report

Project Overview

This project demonstrates the complete data cleaning and exploratory data analysis (EDA) process on a real-world Amazon sales dataset using MySQL. The raw dataset was cleaned by correcting data types, handling missing values, standardizing inconsistent entries, and validating data quality. SQL was then used to analyse sales performance, customer orders, product categories, fulfilment methods, and regional trends to generate actionable business insights.

Tools Used

- MySQL 8.0
- SQL
- Kaggle Amazon Sales Dataset

Dataset Size

- 128,975 records
- 23 cleaned columns

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1. Project Objective

The objective of this project was to clean and analyse a real-world Amazon sales dataset using SQL. The project focused on improving data quality, converting incorrect data types, handling missing values, standardizing inconsistent values, and performing exploratory data analysis (EDA) to uncover business insights. The cleaned dataset will be used to build an interactive Power BI dashboard for decision-making

2. Dataset Overview

Attribute	Details
Dataset Name	Amazon Sales Report
Source	Kaggle
Database	MySQL 8.0
Total Records	128,975
Total Columns	23
Date Range	March 2022 – June 2022
Currency	INR
Purpose	Sales Analysis & Dashboard Development

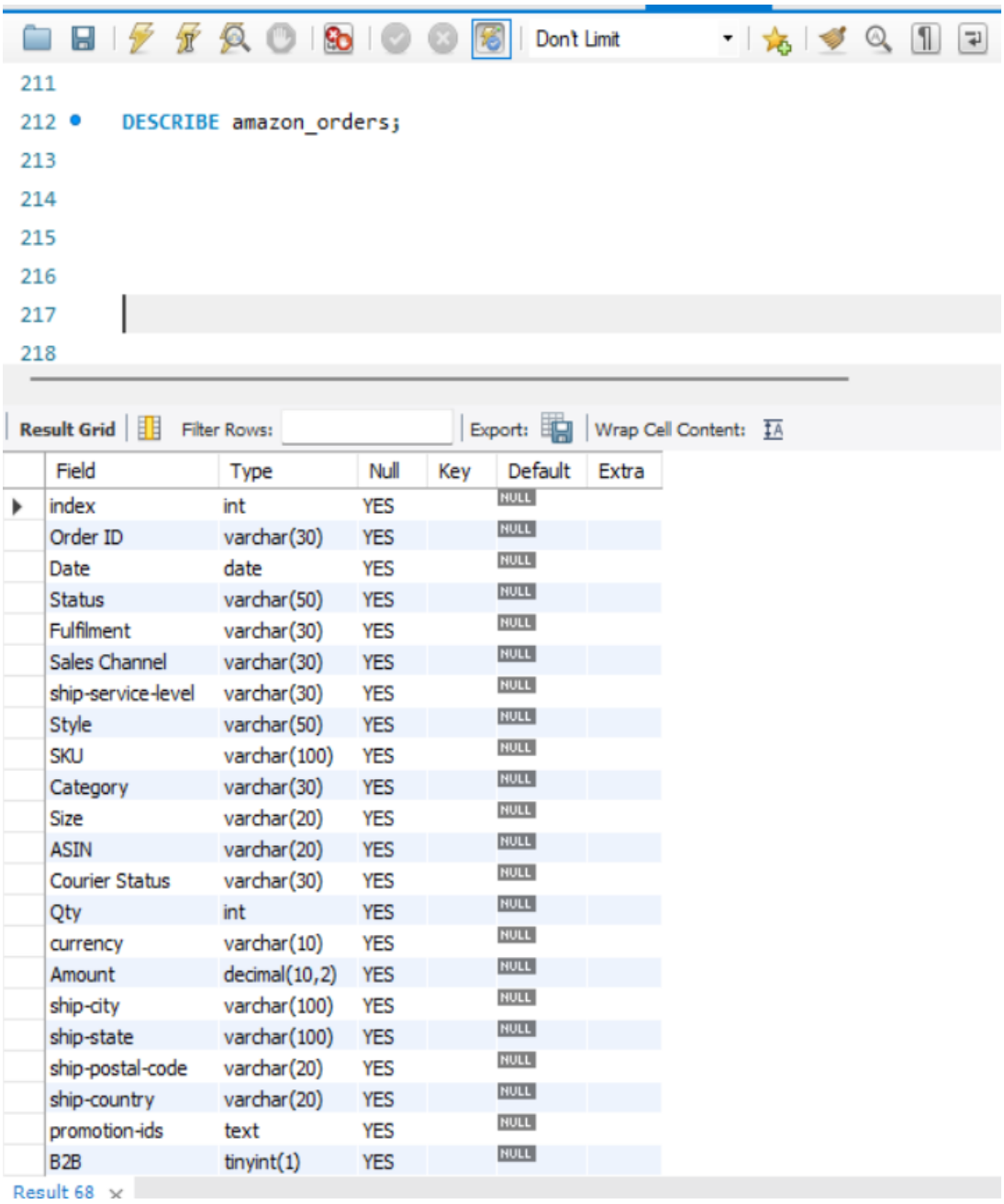
3. Data Cleaning Process

The raw Amazon sales dataset contained inconsistent formatting, missing values, incorrect data types, and duplicate business values. A structured SQL data cleaning process was performed to improve data quality before analysis and visualization.

Data Cleaning Tasks Performed

- Removed the unnecessary **Unnamed: 22** column.
- Converted the **Date** column from text to DATE.
- Converted the **Amount** column from text to DECIMAL (10,2).
- Identified missing values across all columns.
- Checked for duplicate Order IDs.
- Standardized city names using UPPER () and TRIM ().

- Corrected inconsistent state names (e.g., ORISSA → ODISHA, RAJSTHAN → RAJASTHAN).
- Replaced invalid or unknown state values with NULL.
- Validated the cleaned dataset using summary statistics and SQL queries.



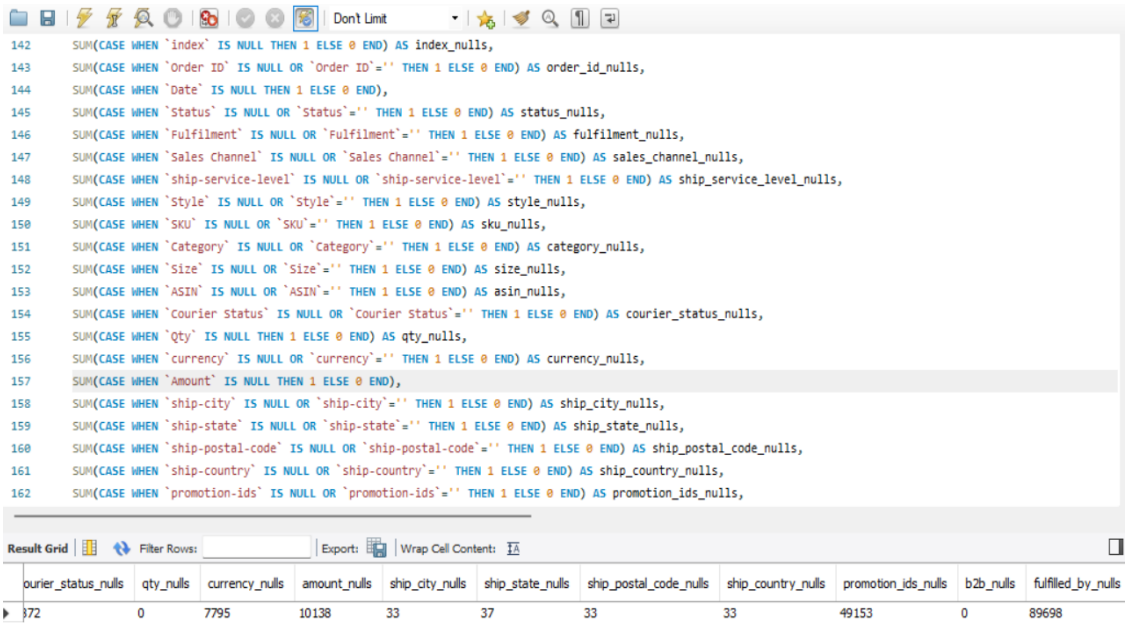
```
211
212 •   DESCRIBE amazon_orders;
213
214
215
216
217
218
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
index	int	YES		NULL	
Order ID	varchar(30)	YES		NULL	
Date	date	YES		NULL	
Status	varchar(50)	YES		NULL	
Fulfilment	varchar(30)	YES		NULL	
Sales Channel	varchar(30)	YES		NULL	
ship-service-level	varchar(30)	YES		NULL	
Style	varchar(50)	YES		NULL	
SKU	varchar(100)	YES		NULL	
Category	varchar(30)	YES		NULL	
Size	varchar(20)	YES		NULL	
ASIN	varchar(20)	YES		NULL	
Courier Status	varchar(30)	YES		NULL	
Qty	int	YES		NULL	
currency	varchar(10)	YES		NULL	
Amount	decimal(10,2)	YES		NULL	
ship-city	varchar(100)	YES		NULL	
ship-state	varchar(100)	YES		NULL	
ship-postal-code	varchar(20)	YES		NULL	
ship-country	varchar(20)	YES		NULL	
promotion-ids	text	YES		NULL	
B2B	tinyint(1)	YES		NULL	

Result 68

FIGURE 1. Table Structure of the Cleaned Amazon Sales Dataset

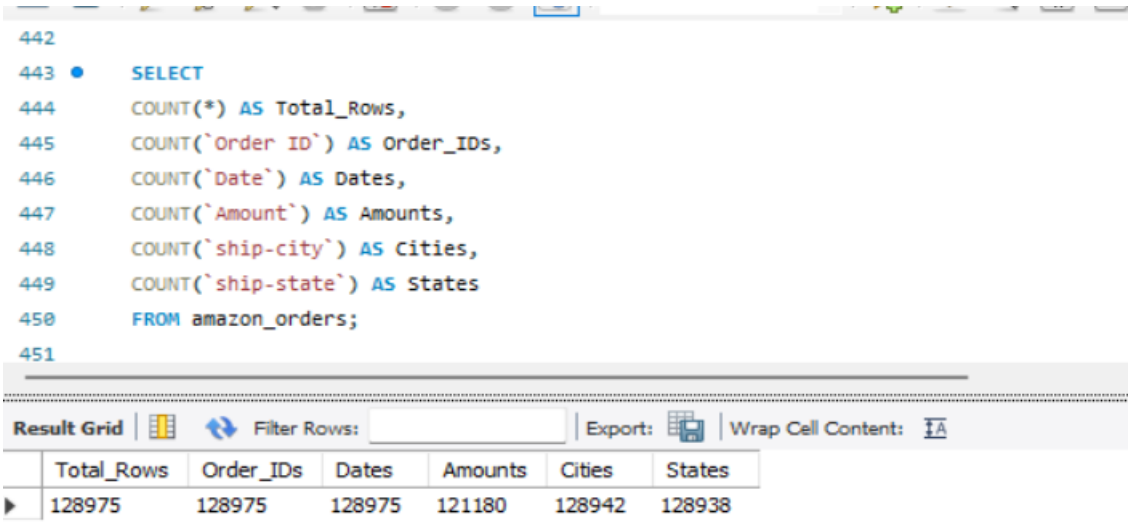


```
142  SUM(CASE WHEN `index` IS NULL THEN 1 ELSE 0 END) AS index_nulls,
143  SUM(CASE WHEN `Order ID` IS NULL OR `Order ID`='' THEN 1 ELSE 0 END) AS order_id_nulls,
144  SUM(CASE WHEN `Date` IS NULL THEN 1 ELSE 0 END),
145  SUM(CASE WHEN `Status` IS NULL OR `Status`='' THEN 1 ELSE 0 END) AS status_nulls,
146  SUM(CASE WHEN `Fulfilment` IS NULL OR `Fulfilment`='' THEN 1 ELSE 0 END) AS fulfilment_nulls,
147  SUM(CASE WHEN `Sales Channel` IS NULL OR `Sales Channel`='' THEN 1 ELSE 0 END) AS sales_channel_nulls,
148  SUM(CASE WHEN `ship-service-level` IS NULL OR `ship-service-level`='' THEN 1 ELSE 0 END) AS ship_service_level_nulls,
149  SUM(CASE WHEN `Style` IS NULL OR `Style`='' THEN 1 ELSE 0 END) AS style_nulls,
150  SUM(CASE WHEN `SKU` IS NULL OR `SKU`='' THEN 1 ELSE 0 END) AS sku_nulls,
151  SUM(CASE WHEN `Category` IS NULL OR `Category`='' THEN 1 ELSE 0 END) AS category_nulls,
152  SUM(CASE WHEN `Size` IS NULL OR `Size`='' THEN 1 ELSE 0 END) AS size_nulls,
153  SUM(CASE WHEN `ASIN` IS NULL OR `ASIN`='' THEN 1 ELSE 0 END) AS asin_nulls,
154  SUM(CASE WHEN `Courier Status` IS NULL OR `Courier Status`='' THEN 1 ELSE 0 END) AS courier_status_nulls,
155  SUM(CASE WHEN `Qty` IS NULL THEN 1 ELSE 0 END) AS qty_nulls,
156  SUM(CASE WHEN `currency` IS NULL OR `currency`='' THEN 1 ELSE 0 END) AS currency_nulls,
157  SUM(CASE WHEN `Amount` IS NULL THEN 1 ELSE 0 END),
158  SUM(CASE WHEN `ship-city` IS NULL OR `ship-city`='' THEN 1 ELSE 0 END) AS ship_city_nulls,
159  SUM(CASE WHEN `ship-state` IS NULL OR `ship-state`='' THEN 1 ELSE 0 END) AS ship_state_nulls,
160  SUM(CASE WHEN `ship-postal-code` IS NULL OR `ship-postal-code`='' THEN 1 ELSE 0 END) AS ship_postal_code_nulls,
161  SUM(CASE WHEN `ship-country` IS NULL OR `ship-country`='' THEN 1 ELSE 0 END) AS ship_country_nulls,
162  SUM(CASE WHEN `promotion-ids` IS NULL OR `promotion-ids`='' THEN 1 ELSE 0 END) AS promotion_ids_nulls,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ourier_status_nulls	qty_nulls	currency_nulls	amount_nulls	ship_city_nulls	ship_state_nulls	ship_postal_code_nulls	ship_country_nulls	promotion_ids_nulls	b2b_nulls	fulfilled_by_nulls
372	0	7795	10138	33	37	33	33	49153	0	89698

FIGURE 2. Missing value Analysis



```
442
443  SELECT
444      COUNT(*) AS Total_Rows,
445      COUNT(`Order ID`) AS Order_IDs,
446      COUNT(`Date`) AS Dates,
447      COUNT(`Amount`) AS Amounts,
448      COUNT(`ship-city`) AS Cities,
449      COUNT(`ship-state`) AS States
450      FROM amazon_orders;
451
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Rows	Order_IDs	Dates	Amounts	Cities	States
128975	128975	128975	121180	128942	128938

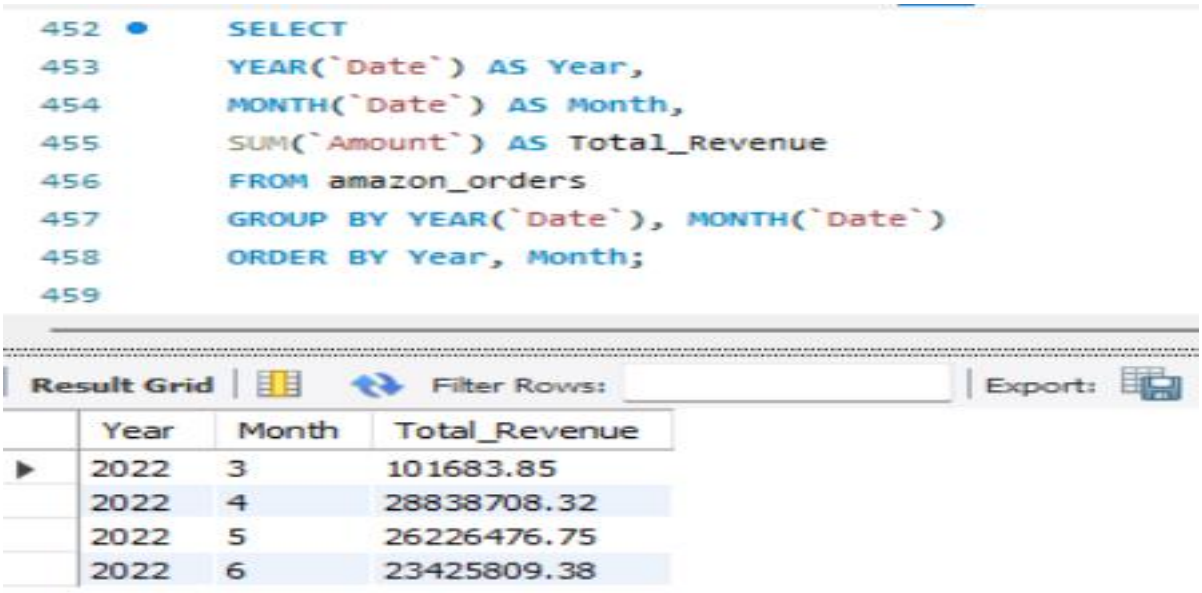
FIGURE 3.DATA VALIDATION

4. SQL Analysis

The cleaned dataset was analysed using SQL aggregate functions and grouping techniques to identify sales trends, product performance, fulfilment efficiency, and regional performance.

Key Analysis Performed

Analysis	Result
Total Revenue	₹78,592,678.30
Total Orders	128,975
Average Order Value	₹648.56
Highest Revenue Month	April 2022
Top Product Category	Set
Top Revenue State	Maharashtra
Top Revenue City	Bengaluru
Main Fulfilment Method	Amazon
Best Selling SKU	JNE3797-KR-L (661 units)

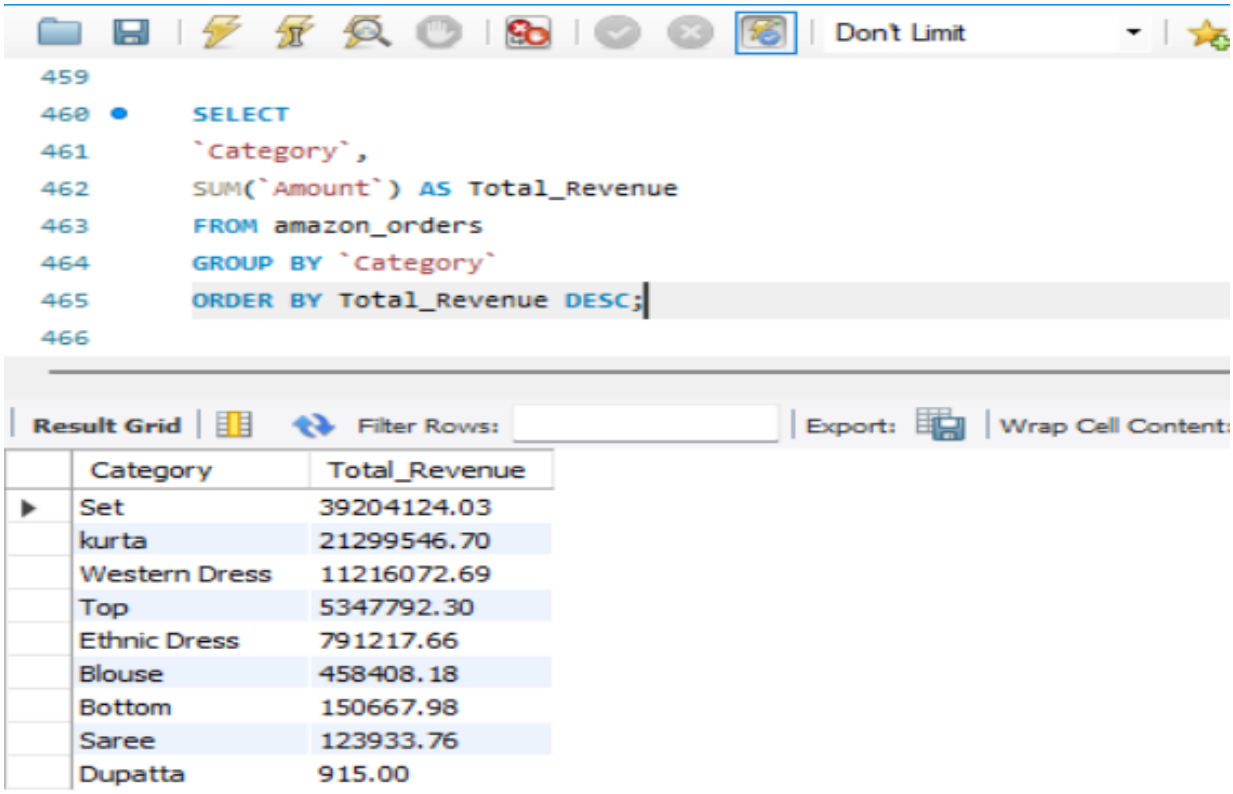


```
452 •   SELECT
453     YEAR(`Date`) AS Year,
454     MONTH(`Date`) AS Month,
455     SUM(`Amount`) AS Total_Revenue
456     FROM amazon_orders
457     GROUP BY YEAR(`Date`), MONTH(`Date`)
458     ORDER BY Year, Month;
459
```

Result Grid | Filter Rows: | Export:

Year	Month	Total_Revenue
2022	3	101683.85
2022	4	28838708.32
2022	5	26226476.75
2022	6	23425809.38

FIGURE 4. Monthly Revenue Analysis



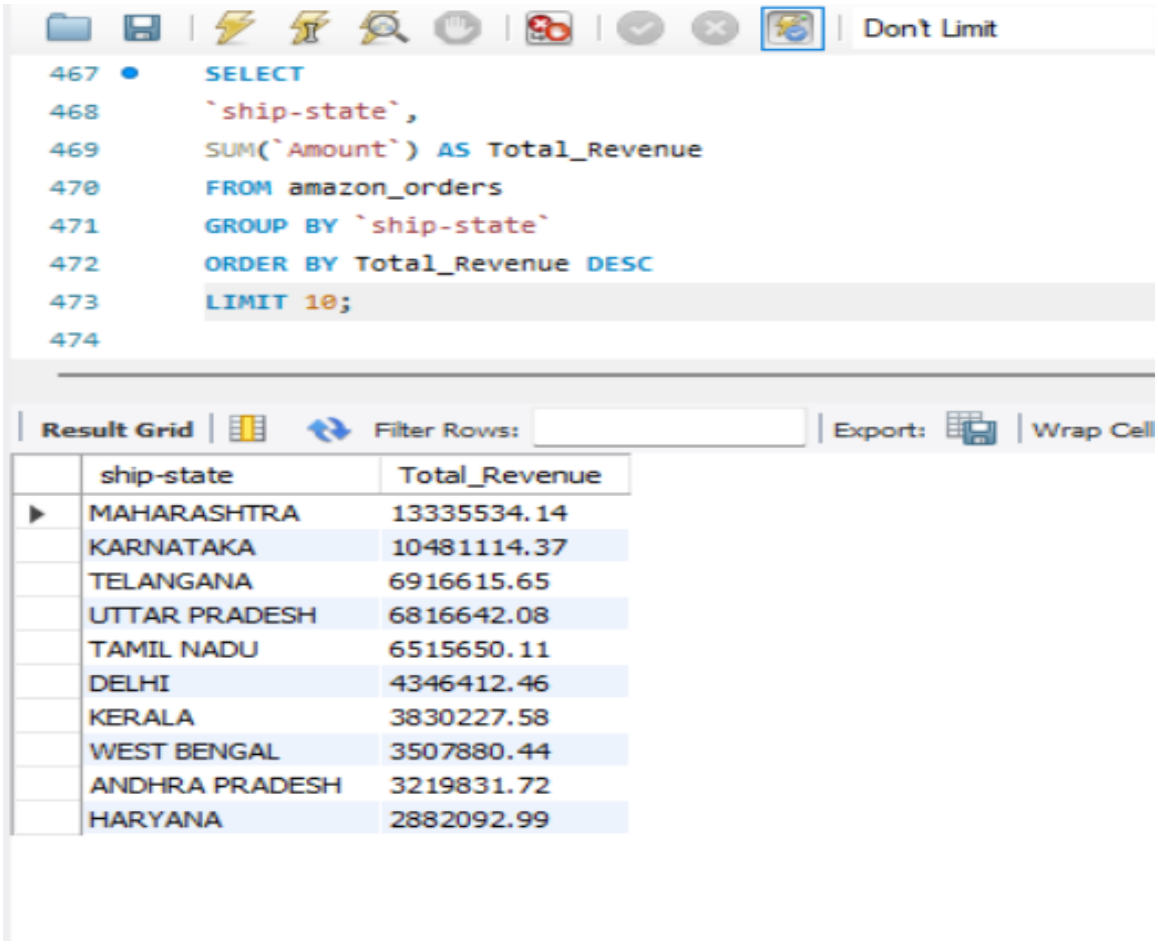
```
459
460 •   SELECT
461     `Category`,
462     SUM(`Amount`) AS Total_Revenue
463     FROM amazon_orders
464     GROUP BY `Category`
465     ORDER BY Total_Revenue DESC;
466
```

Don't Limit

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Category	Total_Revenue
Set	39204124.03
kurta	21299546.70
Western Dress	11216072.69
Top	5347792.30
Ethnic Dress	791217.66
Blouse	458408.18
Bottom	150667.98
Saree	123933.76
Dupatta	915.00

FIGURE 5. Top Product Categories



```
467 •   SELECT
468     `ship-state`,
469     SUM(`Amount`) AS Total_Revenue
470     FROM amazon_orders
471     GROUP BY `ship-state`
472     ORDER BY Total_Revenue DESC
473     LIMIT 10;
474
```

Result Grid | Filter Rows: | Export: | Wrap Cell

ship-state	Total_Revenue
MAHARASHTRA	13335534.14
KARNATAKA	10481114.37
TELANGANA	6916615.65
UTTAR PRADESH	6816642.08
TAMIL NADU	6515650.11
DELHI	4346412.46
KERALA	3830227.58
WEST BENGAL	3507880.44
ANDHRA PRADESH	3219831.72
HARYANA	2882092.99

FIGURE 6. Top 10 States by Revenue

5. Business Insights

Then paste this:

- Amazon generated a **total revenue of ₹78.59 million** from **128,975 orders**.
- **Set** was the highest-performing product category, contributing the largest share of revenue.
- **Maharashtra** recorded the highest sales among all states.
- **Bengaluru** was the top-performing city by revenue.
- **Amazon Fulfilment** accounted for nearly **69%** of total revenue, outperforming Merchant Fulfilment.
- **April 2022** generated the highest monthly revenue during the analysis period.

- Data cleaning significantly improved data quality by correcting inconsistent values and standardizing text fields, making the dataset suitable for reporting and visualization.

6. Conclusion

The Amazon Sales dataset was successfully cleaned, validated, and analysed using MySQL. Data quality issues, including inconsistent text values, incorrect data types, and missing information, were addressed to improve the reliability of the dataset. The SQL analysis identified key business trends related to sales performance, product categories, fulfilment methods, and regional revenue. The cleaned dataset is now ready for visualization in Power BI, where an interactive dashboard will provide deeper business insights for decision-making.